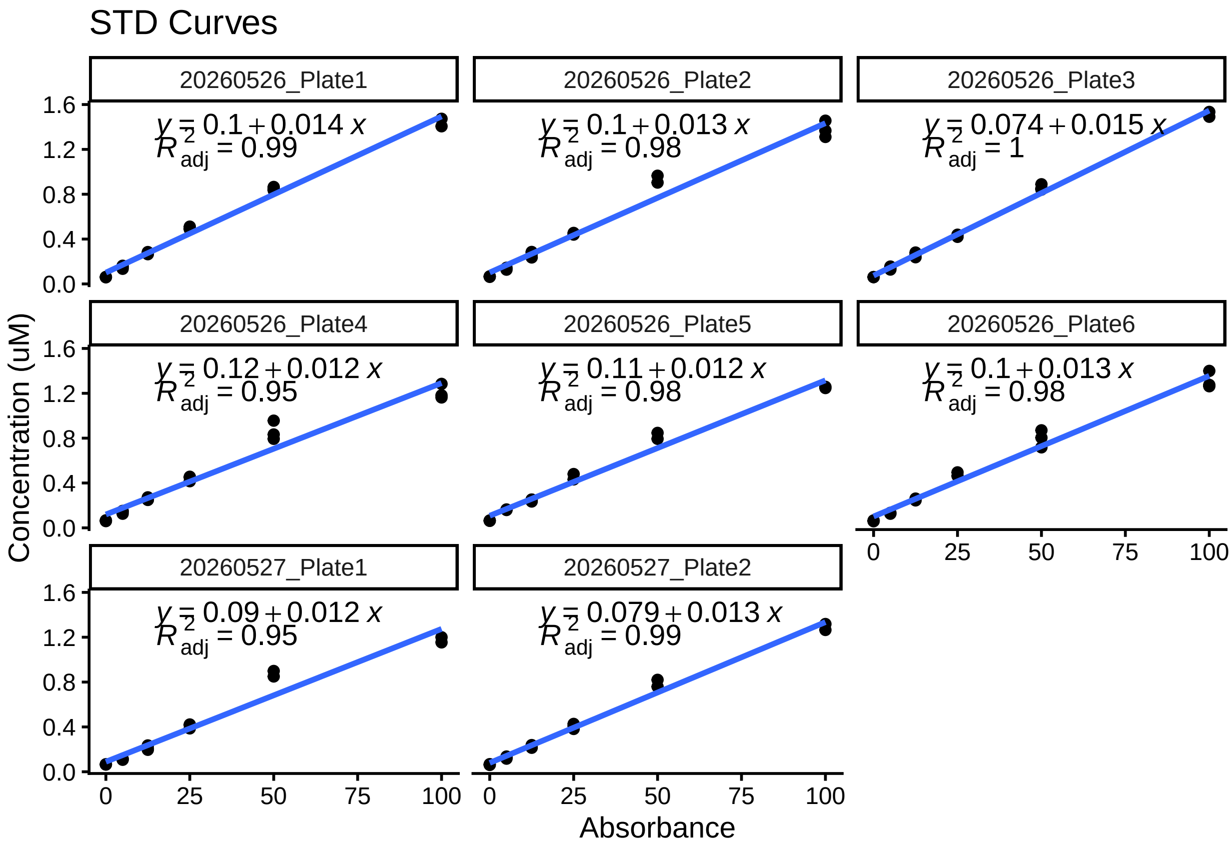
TEMPEST CB: Porewater Sulfide

March-May 2026 Samples

2026-07-28

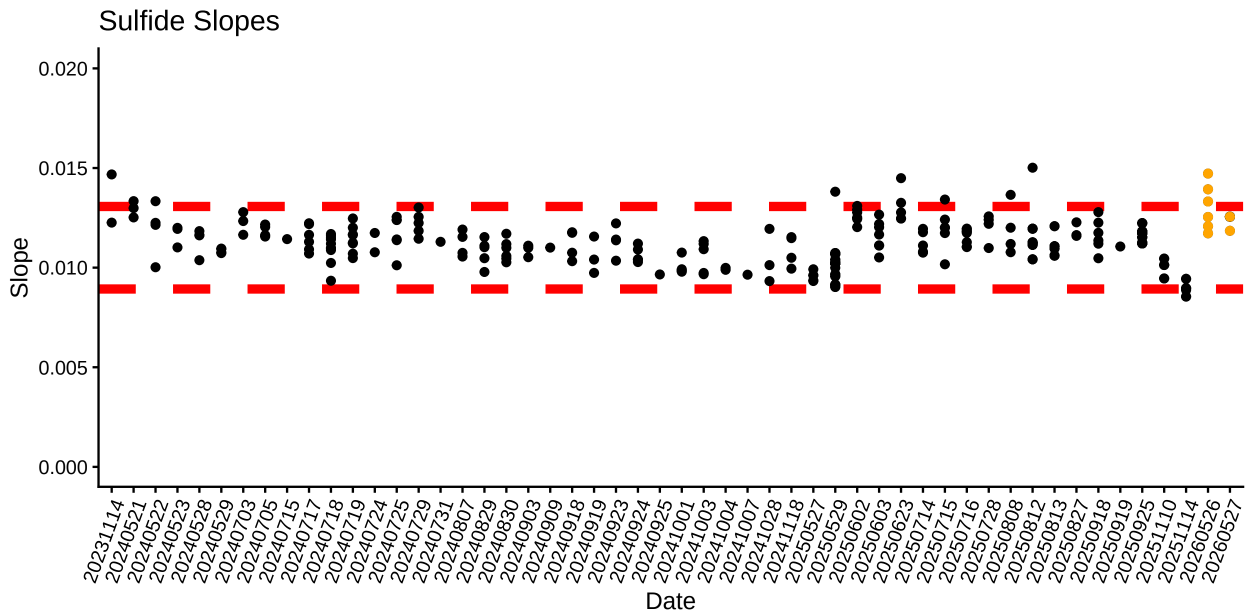
Run Information

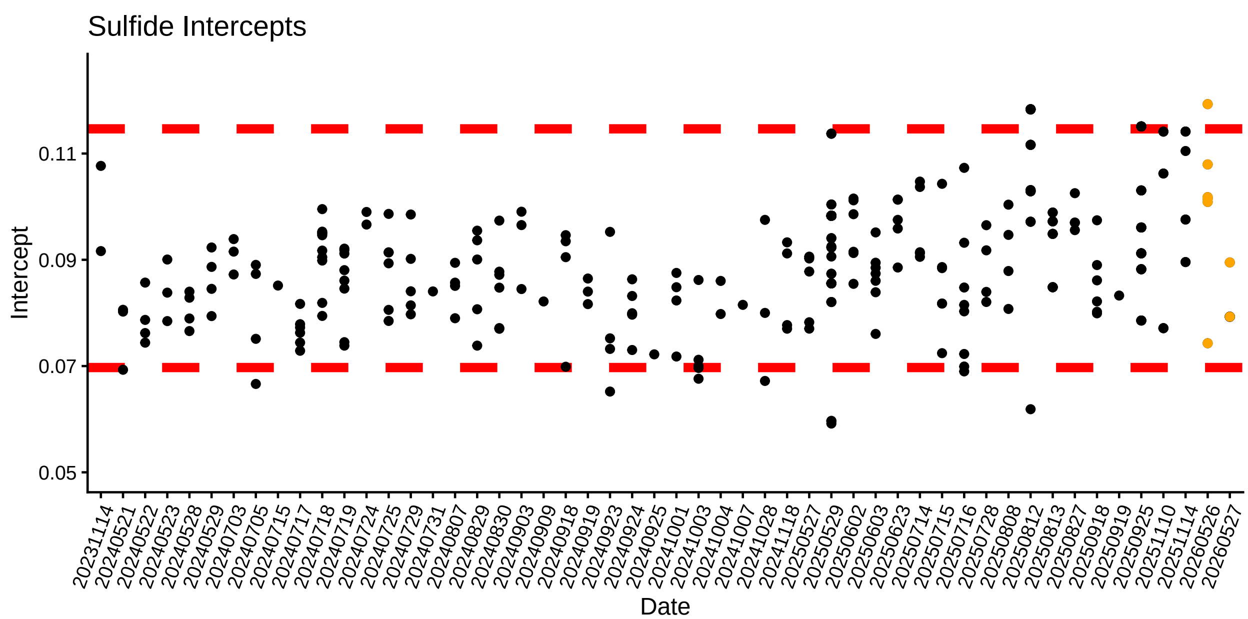
```
###things that need to be changed
Date_Run = "20260526-20260527"
Month = "March-May"
Year = "2026"
Sample_Month_Year = "202605"
Run_by = "Zoe Read"  #Instrument user
Script_run_by = "Zoe Read" #Code user
Project = "COMPASS"

Run_notes="
Missing TMP_FW_PW_B4_20260410_15CM and May 2026 plots from metadata.
Used 20260527_Plate2 std curve.
3/16 Matrix checks were bad.
3/8 dups were bad, probably because the sulfide values were so low.
5/8 spikes were bad" #any notes from run

#Stds that should be excluded
# stds_to_remove<-data.frame(Plate=c("Plate6"),IDs=c("Std 3"))
stds_to_remove<-NA
```



STD Curves
20260526_Plate1
20260526_Plate2
20260526_Plate3
20260526_Plate4
20260526_Plate5
20260526_Plate6
20260527_Plate1
20260527_Plate2
$y = 0.1 + 0.014\,x$ $R^2_{adj} = 0.99$
$y = 0.1 + 0.013\,x$ $R^2_{adj} = 0.98$
$y = 0.074 + 0.015\,x$ $R^2_{adj} = 1$
$y = 0.12 + 0.012\,x$ $R^2_{adj} = 0.95$
$y = 0.11 + 0.012\,x$ $R^2_{adj} = 0.98$
$y = 0.1 + 0.013\,x$ $R^2_{adj} = 0.98$
$y = 0.09 + 0.012\,x$ $R^2_{adj} = 0.95$
$y = 0.079 + 0.013\,x$ $R^2_{adj} = 0.99$
Concentration (uM)
Absorbance
0.0
0.4
0.8
1.2
1.6
0
25
50
75
100

Checking STD Data against QAQC file





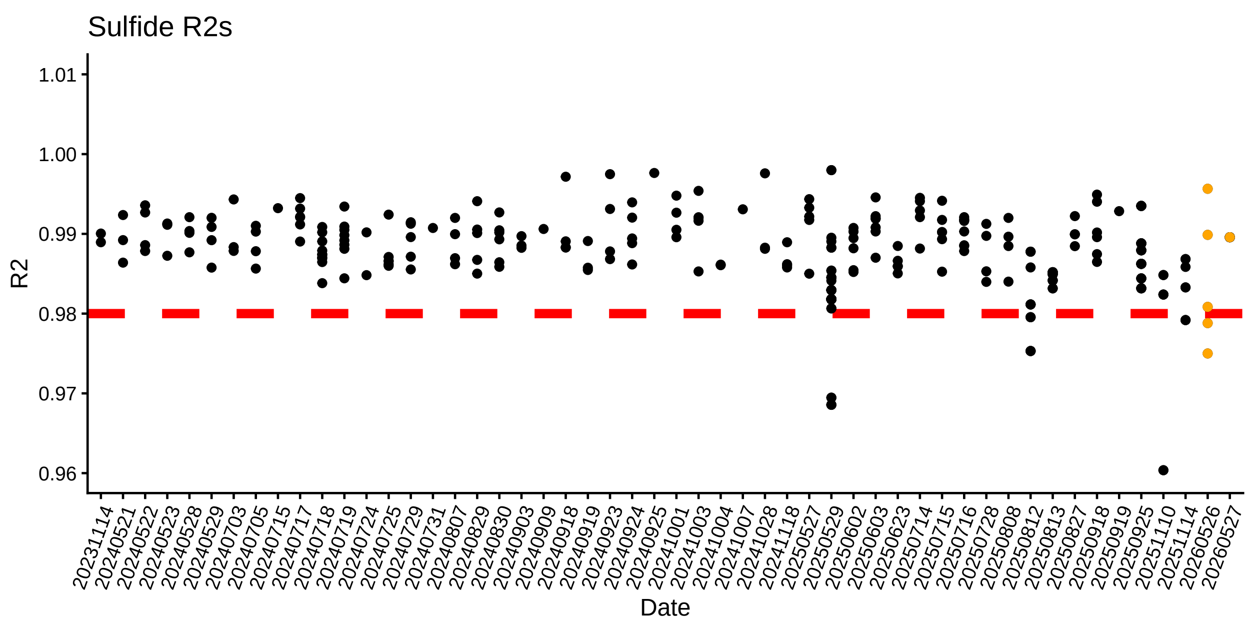
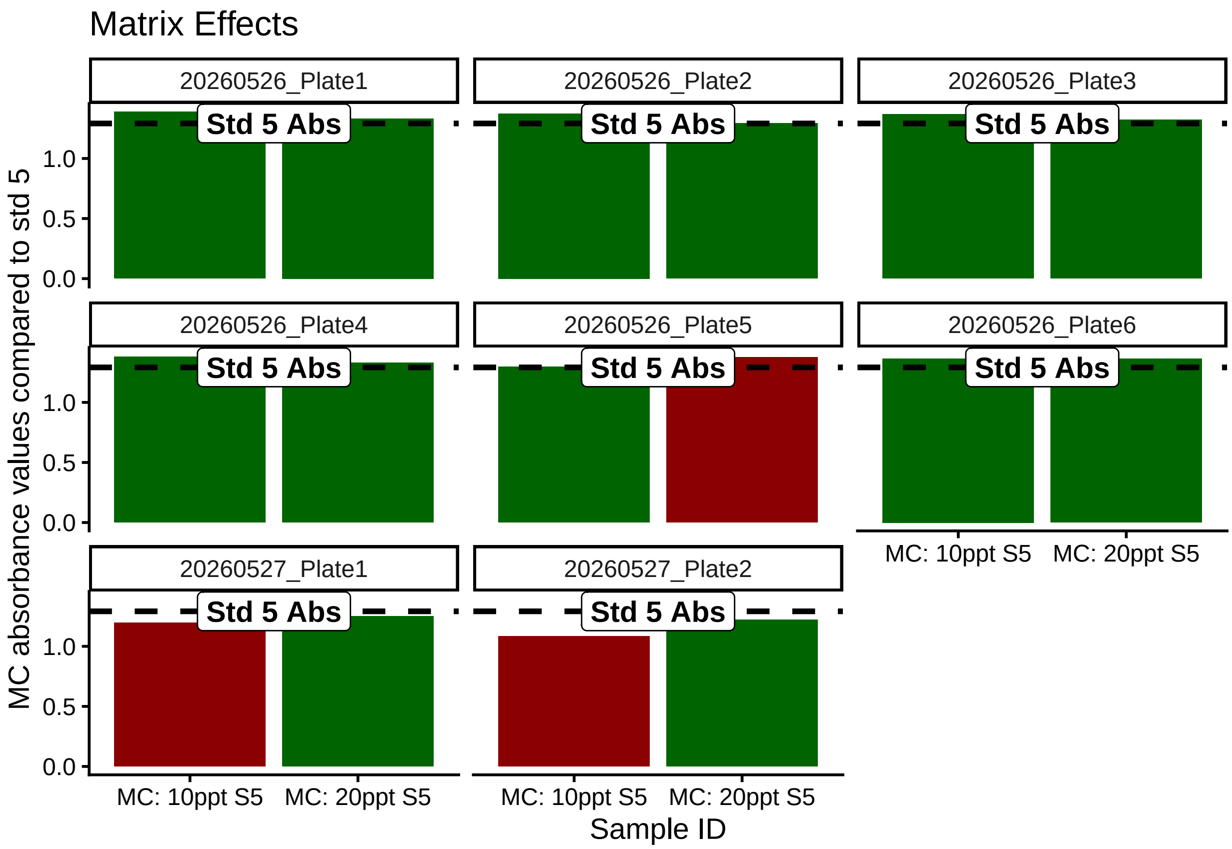
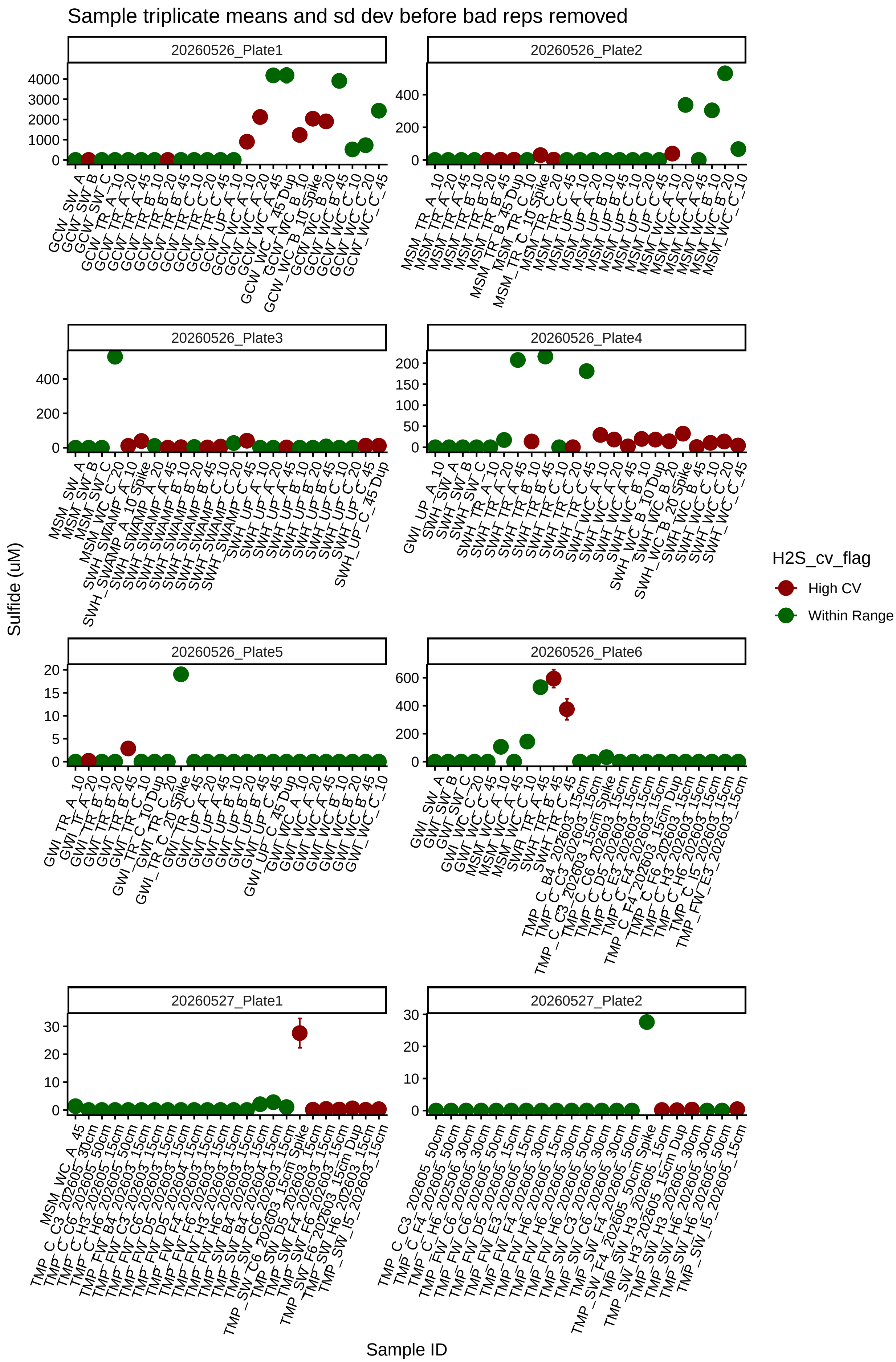


Table 1: Best std curve to use:

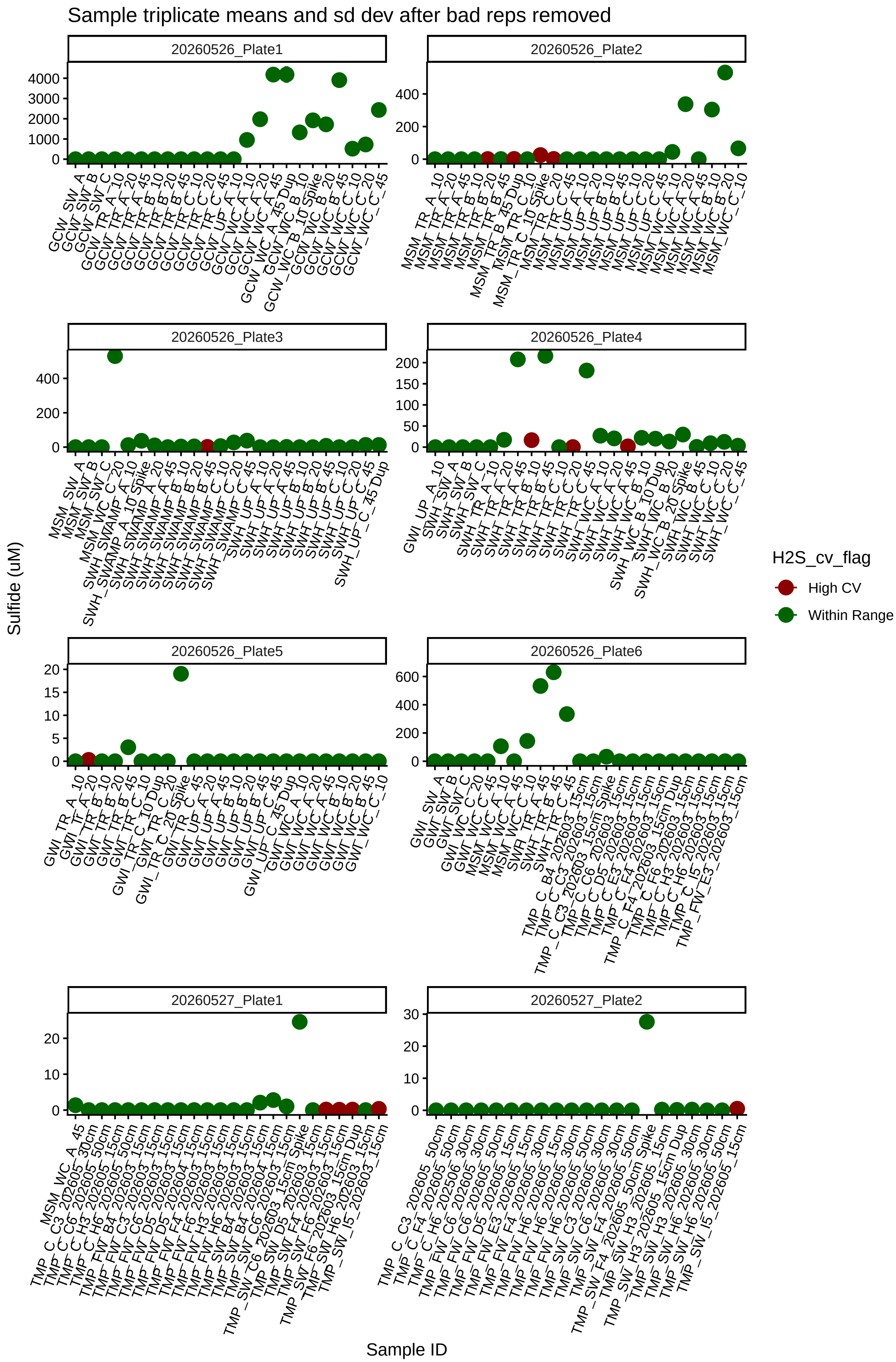
Date	Project	R2	Slope	Intercept	Top_STD	Plate
20260527	COMPASS	0.9895755	0.0125559	0.0792854	100	20260527_Plate2

Matrix Check QAQC

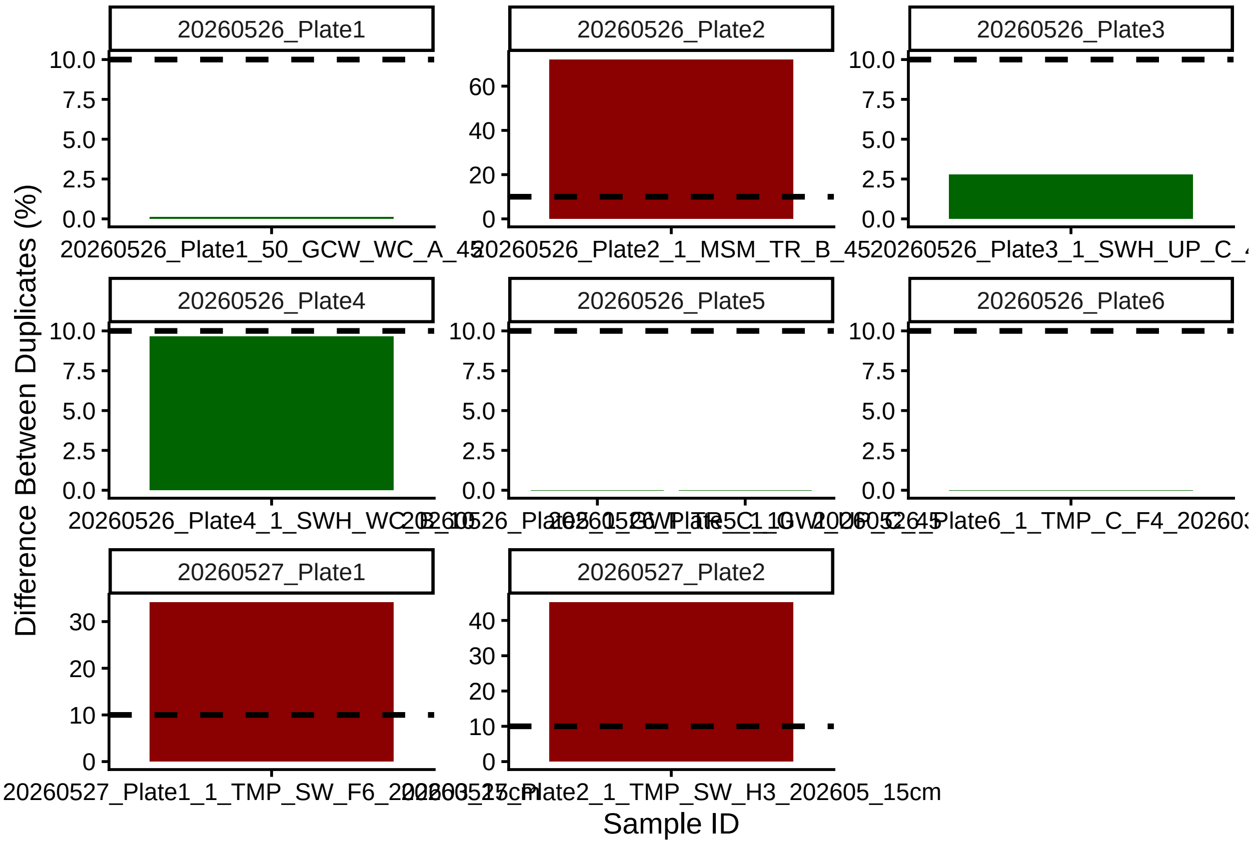



Sample triplicate means and sd dev before bad reps removed
20260526_Plate1
20260526_Plate2
20260526_Plate3
20260526_Plate4
20260526_Plate5
20260526_Plate6
20260527_Plate1
20260527_Plate2
Sulfide (uM)
Sample ID
H2S_cv_flag
High CV
Within Range

Remove bad reps

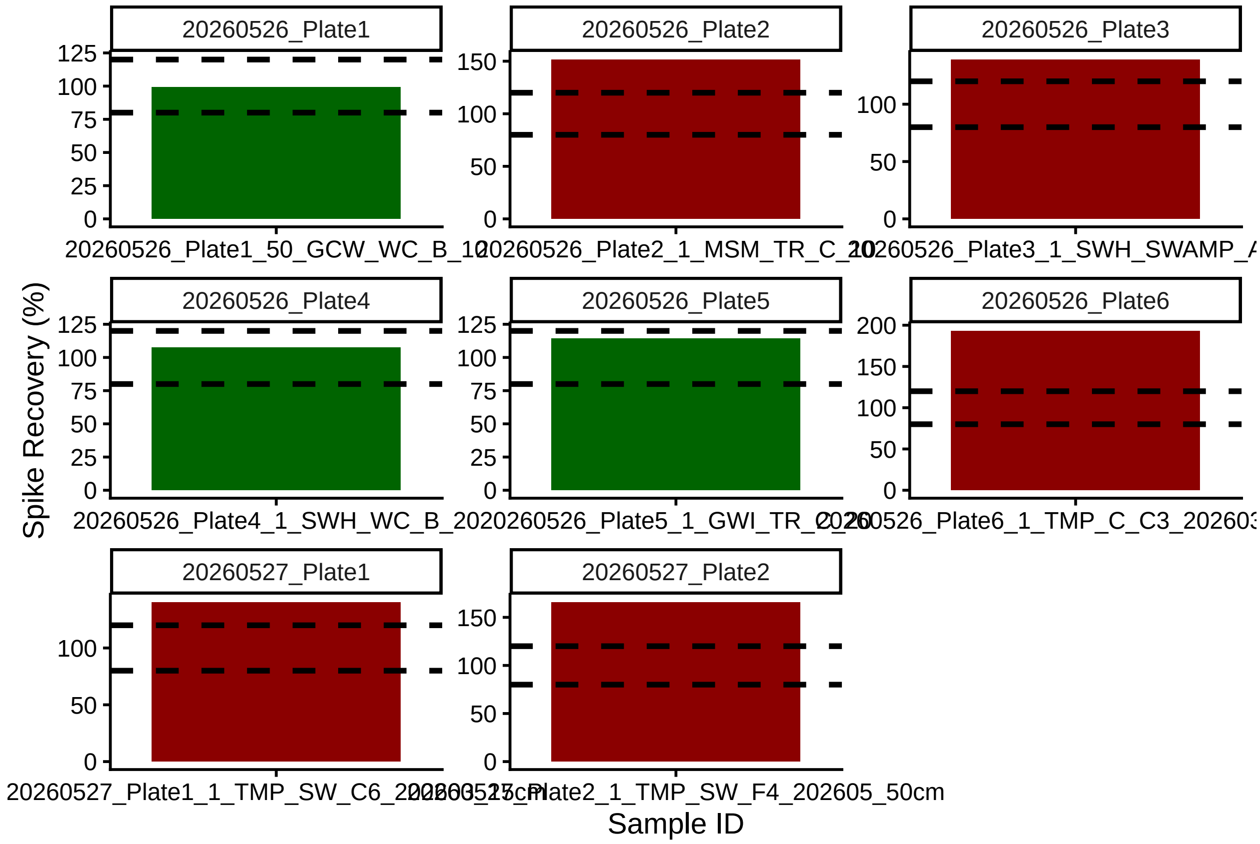


Check the dups for QAQC



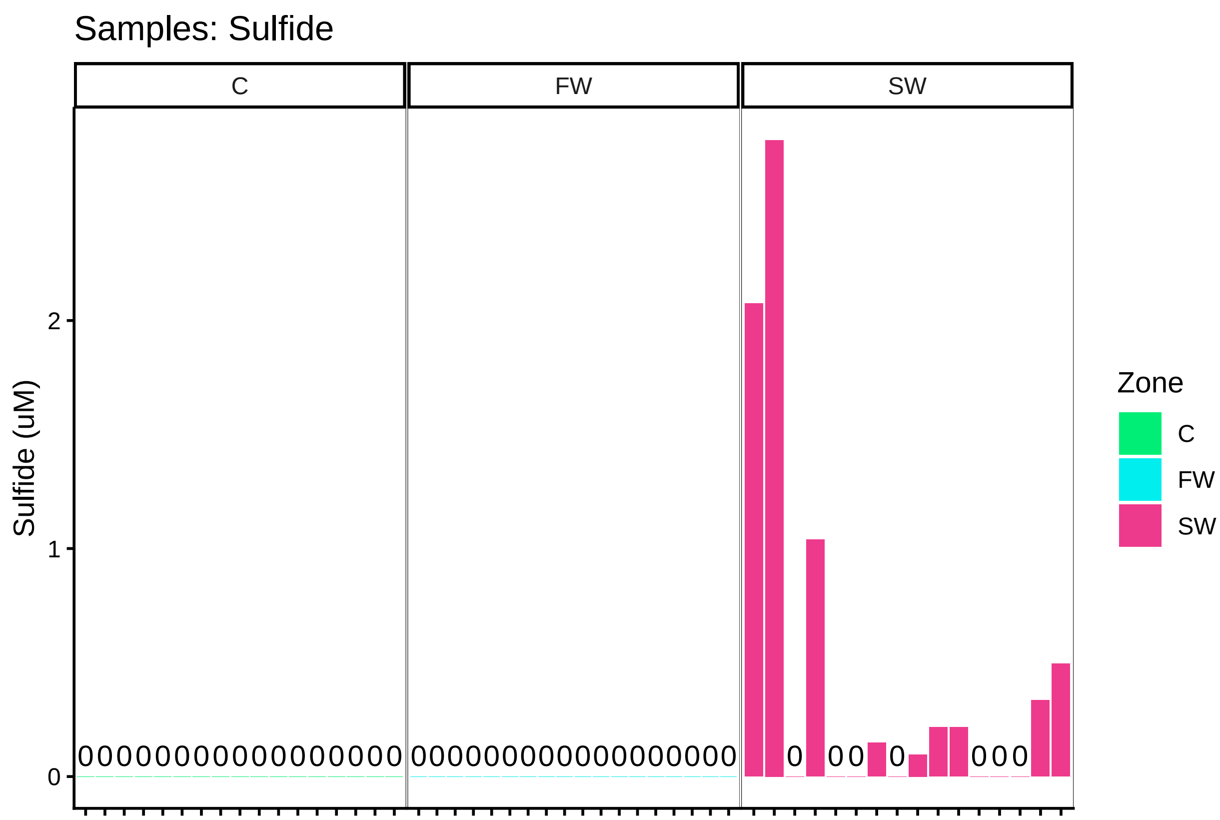

```
## [1] ">60% of Duplicates are within <10%"
```

Check the spks for QAQC




```
## [1] "<60% of Spikes are out of range - REASSESS"

## ***All sample IDs are present in metadata.***
```

Visualize Data by Plot



###END